Datatypes

Data : Information

- Constant
- Variable

Bio-Data

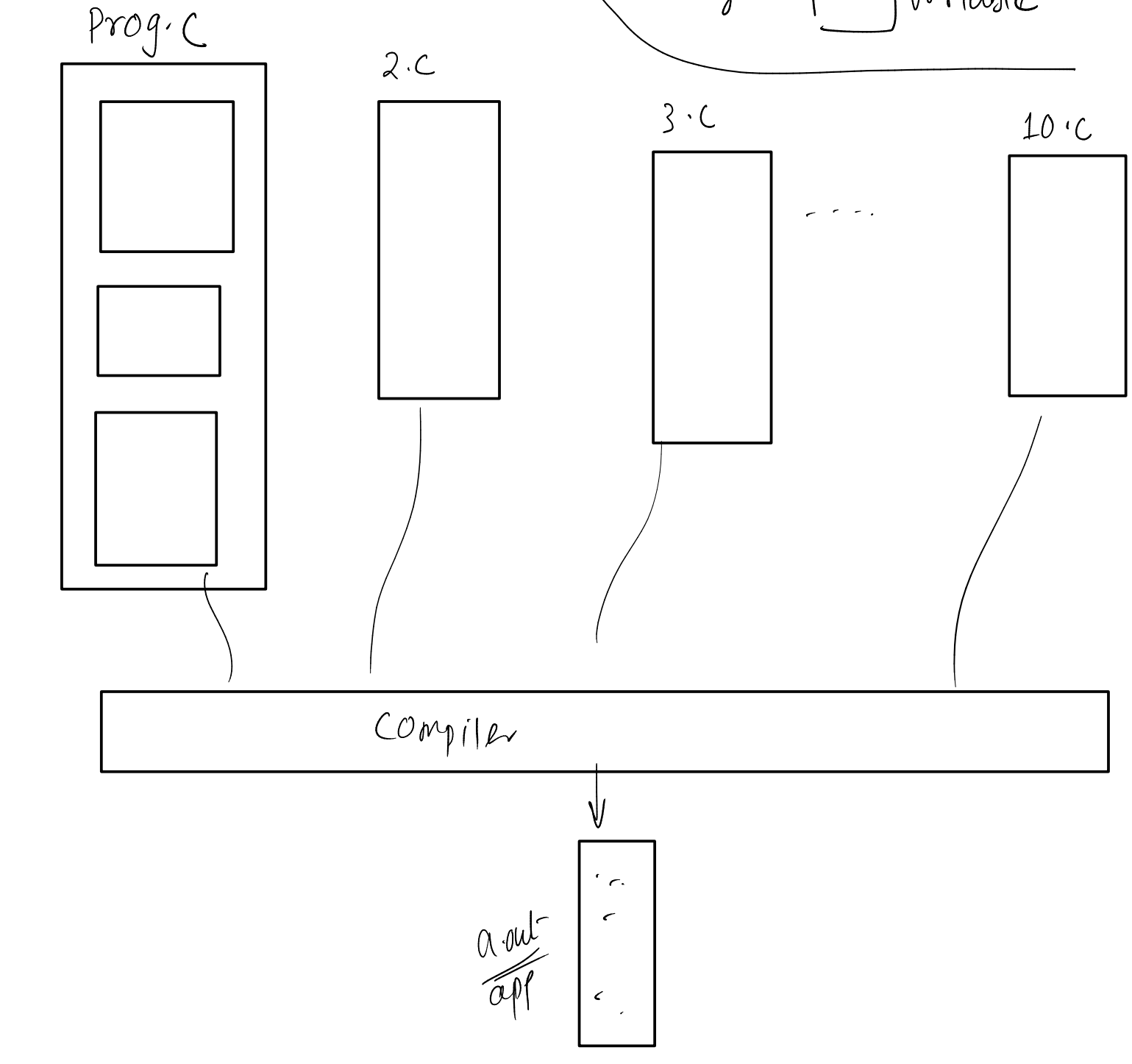
Name: ABC constant

D.O.B: ..... Constant

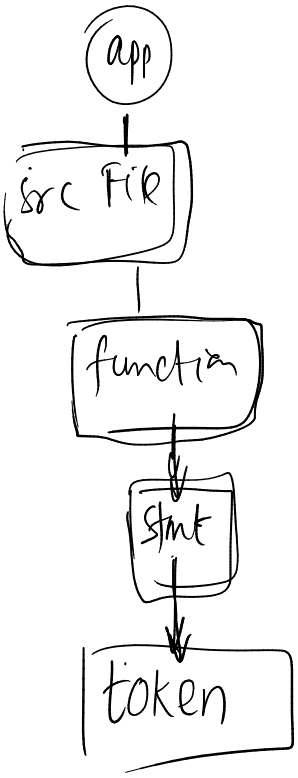
Y.O.P:

Qualifn — variable

Age: variable



- An app is created from one (or many) source files.
- A source file contains one/many function Block.
- A function contains one/many statements.
  → a Stmt contains one or many tokens.



Token

→ Smallest unit of a program

- Keyword: Reserved word / meaningful to compiler.
- Identifier: The words not known to the compiler.
  → variable, function
  Hence MUST BE Declared.
- Operator: A symbol, designated to perform a specific operation
- Delimiter/separators: , ; (...
- Constants: 'A' '*' '5' '_' 't' => character constant

  567, 0X123, 056, -234 Integer constant

  5.67, -0.05, 3.45 @+2 Real constant / floating point constant

  "5.67", "vector", "5", "#" : string constant

info@vectorindia.org

printf : Identifier, must be declared.

```
#include<stdio.h>
=
=
```

```
=
=
int printf(const char*, ...);
```

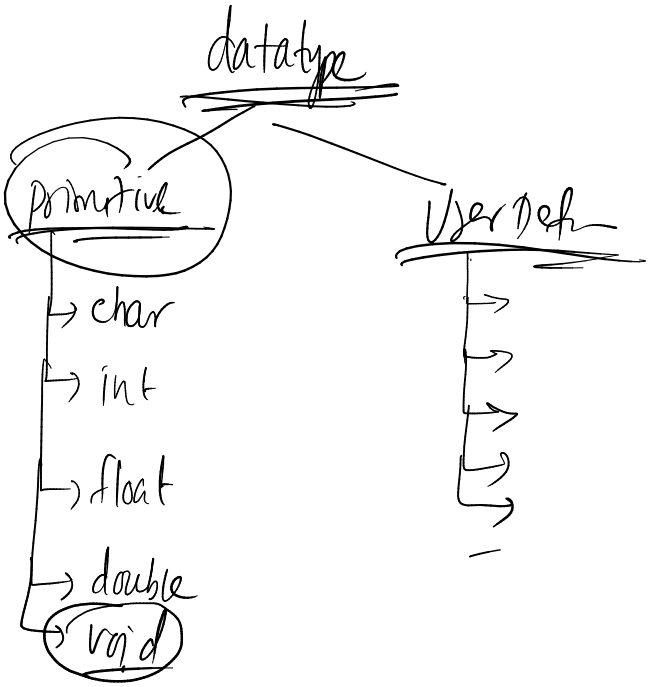
Why main is not declared ?

---

variable

datatype var_name;

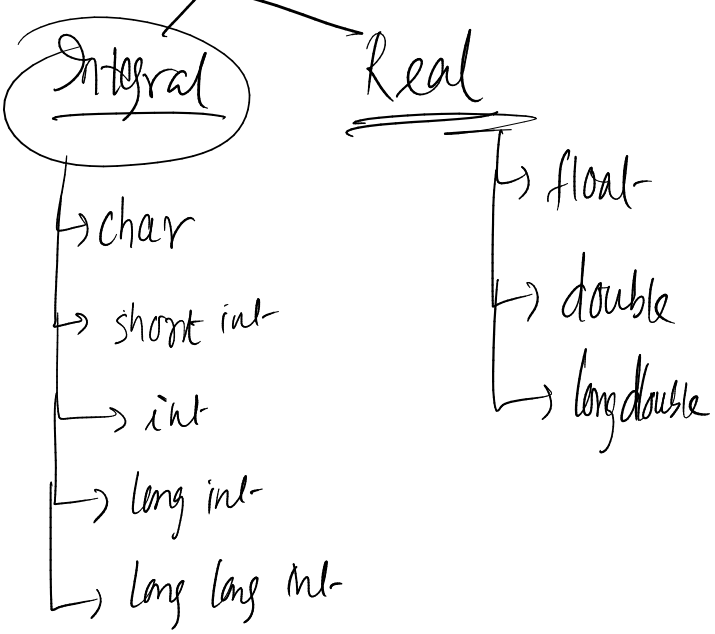
```
main(){
  void x; //Error
  void *y; //Generic Pointer
}
```




```
void f1(void)
{
=
=
=
}
```

This function f1, receives NO-Data

f1 returns NO-data



ASCII

'a' => 01100001 => 97

'b' 98

```
char x = 'a';
         97
```

x

0	1	1	0	0	0	0	1

← 1 byte →

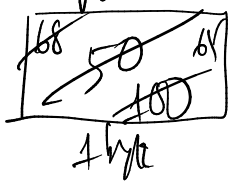
```
char y = 97;
```

y

7	6	5	4	3	2	1	0
0	1	1	0	0	0	0	1

```
Putchar(x); // a
putch(y); // a
```

In 'C' compiler, the character constants are treated as Integer constants (their ASCII equivalent)



Var

~~50~~

~~100~~

~~68~~

65

'A' = 65

'B' = 66

.

.

.

'a' = 97

'b' = 98

'0' => 48

'1' => 49

'2' = 50

Putchar()

Puts("Hello");

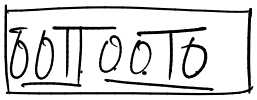
printf

with a format ~~specifier~~

Format specifier %c %s %d

char var = 50;

var



Pf("%d", var); // 50

"%c", var); // 2

"%x", var); // 32

%o // 62